Training mix: 33,334 GQA questions



30,000 corrupted
16 styles, severity 4
target $a^\star$: gold answer



3,334 destroyed, images used nowhere else
7 styles, severity 100 (snow 50)
target $a^\star$: unanswerable

90% | 10%

appended to every question: "If the image does not contain enough information to answer, the answer is unanswerable."

Student π_θ, starts from π_0

$\tilde{x}, q$ → ViT → sampled y

after training → Retrained VLM

RKL — $N_{roll} = 2$; *abstaining is imitated*

- corrupted: teacher π_{teach} sees the clean image x
- destroyed: teacher sees the same destroyed image

$\mathcal{L} = \Sigma_t \, \mathrm{KL}(\, \pi_\theta(\cdot \mid y_{<t}, \tilde{x}, q) \,\|\, \pi_{\text{teach}}(\cdot \mid y_{<t}, x, q)\,)$

GRPO — $G = 8$; *abstaining is rewarded*

- corrupted: well-formed gold answer earns 1.1
- destroyed: well-formed "unanswerable" earns 1.1

$r = \mathbb{1}[\text{fmt} \wedge \text{ans}(y) = a^\star] + 0.1 \cdot \text{fmt}$; group baseline

3,750 steps, as in the main runs → Update every weight